

Developing Autonomy-by-design Recommendation Algorithms to Empower User Autonomy and Self-realisation

Yushi Yang
Word counts: 2496

1 Motivation

Modern algorithms can influence the behaviour of users through persuasive tools such as tailoring, tunneling, and suggestions, which may exploit the psychological weaknesses of users (Fogg, 2003). If these algorithms aren't adequately governed, they can gradually manipulate people into achieving goals that deviate from their interests and violate their decision-making rights (Botes, 2022). Immersion in such technologies can influence people's perception of reality and even their basic value judgments (Susser et al., 2019). At worst, it could lead to the collapse of Kantian principles of autonomy (Kleingeld, 2017), where humans are treated as a means, rather than an end, and democratic principles are at stake.

For now, the most powerful persuasive technology driving personal transformation is the recommendation systems (RS) (Michael Schrage, 2020). With a tap or swipe, recommenders turn 'next moments' into graspable opportunities with tailored options. They curate news feeds, power job and dating sites, and promote targeted advertisements. By controlling how options are presented and information is exchanged, they can reconstruct personal preferences and shape the public opinion (Milano et al., 2020a). In particular, personalised recommendations can interfere with user autonomy in several ways, by nudging them in a particular direction, trying to 'addict' them to certain types of content, or restricting their options (Djeffal et al., 2022). Several real-life examples illustrate these points, such as how people make impulsive online purchases they later regret (Aragoncillo Caballero and Orús, 2018), how they complain about social media wasting their time (Hiniker et al., 2016), or how a teenage girl committed suicide after falling into a dark hole of suicidal content (Dirnhuber, 2019). Such persuasive effects of RS range from small nudges to premeditated manipulations of individuals (Dumbrava, 2018). The notorious Cambridge Analytica scandal, gathered data about 87 million Facebook users to create psychologically targeted advertisements to influence people's voting preferences in the 2016 US presidential election (Hinds et al., 2020). This shows how deliberately crafted recommendations can undermine citizens' ability to exercise democratic self-determination.

The concept of autonomy, according to Kant, involves (i) the right to make one's own decisions without interference from others; (ii) the capacity to make such decisions through one's own independence of mind and after personal reflection (Kleingeld, 2017). Accordingly, one could argue that RS's damage to autonomy is inevitable. If people had complete

autonomy over RS, they would need to resist any initial recommendations that diverge from their browsing intent (to keep ‘independence of mind’), and all future recommendations should be solely based on users’ previous preferences (with ‘no external interference’). This is not practical given the fragility of human nature, and simply doesn’t fit with most RS platforms’ engagement-based profit models (Calvo et al., 2020). Given that RS’s shaping power on autonomy is inevitable, I argue that RS designers should direct its shaping power in a positive direction. Thus, I propose ‘self-realisation’ as an alternative design goal to safeguard user autonomy. This design goal echos the view that RS are essentially an ‘augmented introspection’ of today’s humans, reflecting and magnifying ‘who we are’ as ‘we are what we choose’. In this perspective, the future for RS lies in empowering individual users to discover themselves and drive meaningful self-awareness (Michael Schrage, 2020).

Self-realisation, in Western philosophical terms, is the “fulfillment by oneself of the possibilities of one’s character or personality” (Evans, 1998). In the context of personalisation, it can be conceptualised as helping users to explore, understand, and develop their unique tastes and preferences (Knijnenburg et al., 2016). RS platforms may be more commercially successful if they provide users with self-discovery opportunities (Knijnenburg et al., 2016). Autonomy-wise, a person who has achieved self-realisation is theoretically capable of making informed decisions without external influence (John Walter Bordages, 1989). Promoting self-realisation on a collective level can also lead to a more conscious society as self-realised individuals are generally more independent, resourceful, and accepting (Benn, 1988). As a result, self-realisation provides a good middle ground between balancing platform interests and serving the common good for individuals and society, resolving the conflicting goals between different stakeholders in RS.

The research proposal is structured as follows. Section 2 examines how promoting self-realisation on RS can practically mitigate autonomy risks. Section 3 outlines the specific research questions. Section 4 presents the proposal’s main methodologies, including a framework for promoting self-realisation in RS and the algorithm design. Section 5 examines the limitations of this research.

2 Autonomy to Self-realisation: An Enhancement

Based on the literature, the erosion of user autonomy in RS is mainly caused by four factors:

- **Conformity Loop:** RS have been shown to have persuasive properties, where over time, users are more likely to agree with a recommender’s ratings (Cosley et al., 2003) and to follow its recommendations (Gretzel and Fesenmaier, 2006), thus unwittingly making themselves more ‘fit in’ with the system.
- **Addictive Junk Food:** Many RS have a tendency to recommend items that are easily to be consumed or liked (‘Junk Food’) in order to maximise user engagement (Boyd, 2010). This exploits a human tendency called temporal discounting (Doyle, 2012), where people choose instant gratification (e.g. funny memes) over long-term educational experience in pursuit of short-term pleasure.

- **Filter Bubble:** RS may impose a myopic view of the world on users by recommending homogeneous content and isolating them from diverse viewpoints and experiences (Pariser, 2012). The Filter Bubble phenomenon in RS has been validated (Nguyen et al., 2014), although to a lesser extent than originally claimed.
- **Social Manipulation:** RS may be manipulated by particularly active groups of users, whose interactions with the system generate intense positive feedback, thus driving up the system’s rate of recommendations for specific items (Milano et al., 2020b). For example, recent years have seen RS for news feeds and social networks become venues for targeted political propaganda (Hinds et al., 2020).

Promoting self-realisation has potential to address the autonomy risks outlined above. By encouraging users to explore their underdeveloped tastes, the filter bubble will burst; as users become more aware of their unique interests, the conformity loop will be broken; as we design algorithms that prioritise individual preferences over majority tastes, the likelihood of promoting unwanted ‘junk food’ and social manipulation will be reduced.

3 Research Questions

This research aims to develop autonomy-by-design algorithms to promote users’ self-realisation in RS. Translating a value like self-realisation into design requirements involves three steps: conceptualising the value, translating a general value into metrics, and specifying the design requirements (Natali Helberger, 2020). Following this path, I structure the following research questions (RQ):

1. How to design a RS framework to promote self-realisation and respect user autonomy?
2. What quantifiable metrics are useful in measuring the effects of self-realisation and how to formulate them?
3. How to incorporate these metrics into the recommendation algorithms and how to design algorithms to implement the framework as proposed in RQ1?
4. How to evaluate the effectiveness of the proposed algorithms?

4 Research Design

Section 4.1 proposes the framework for promoting self-realisation in RS, including relevant metrics and algorithm designs. Section 4.2 presents the potential datasets and evaluation methods for the research.

4.1 Framework for Promoting Self-realisation in RS

As a response to RQ1, I propose the following steps for promoting self-realisation in RS: (i) Gain a deeper understanding of user preferences (RQ2); (ii) Give users more control over

content selection; (iii) Help users discover their underdeveloped preferences (RQ3). The focus of this framework is on recommending items that contain stories or ideas, such as articles, books, and videos, that may have a greater impact on people’s minds than just products on e-commerce sites. Thus, I will use the terms of recommended ‘items’ and ‘content’ interchangeably going forward.

4.1.1 Gain a deeper understanding of user preferences

Meta-preference Learning: Traditional engagement metrics (clicks, browsing time) and item categories (e.g. ‘comedy’, ‘thriller’) do not adequately reflect user preferences in RS. This is because user preferences are often fleeting, constructed on the fly, and prone to distortions (Amatriain et al., 2009). So when using an iterative machine learning algorithm, it is difficult to determine whether the algorithm has learned about its users, whether the users have changed, or if the system has taught users to behave in ways to maximise the objective function (Ashton and Franklin, 2022a). As a better indicator, users’ meta-preferences, i.e. their preferences over preferences, are more stable indicators to distinguish between welcoming preference shifts from unwelcoming or even manipulative ones (Ashton and Franklin, 2022b). For an example of meta-preferences, a person might enjoy a certain film, but simultaneously wish they didn’t, so we might describe it as a guilty pleasure (see Table 1). Similarly, they might dislike exercise and dislike that about themselves (a guilty-hatred). Further, they may also like that they like certain things (righteous-pleasure) and like that they dislike other things (righteous-hatred). To understand the governance structure of users’ preferences, it is crucial to understand their meta-preferences.

Meta Preference	Preferences	
	Likes	Dislikes
Likes	‘Righteous-pleasure’	‘Righteous-hatred’
Dislikes	‘Guilty-pleasure’	‘Guilty-hatred’

Table 1: Meta preferences are the preferences over preferences (Ashton and Franklin, 2022b)

Measure the Nutrition Levels of Items: How ‘nutritious’ (i.e. informative or meaningful) a piece of content is perceived by a user is an important indicator to distinguish their guilt-pleasure (GP), righteous-pleasure (RP), or guilt-hatred (GH) preferences. Although users perceive content differently, there are some criteria that can indicate a content’s general ‘nutrition’ value, such as its educational quality, technicality, whether the topic is generally regarded as good, source credibility, etc. One way to compute these metrics is to input metric-related features, such as text complexity, readability, tone, and sentiment (derived from Natural Language Processing models), as well as topic categories and tags associated with the content, into a deep-learning model such as random forests or neural networks to learn the metrics levels (Kurdi, 2019). Even if the content is a video, one can still use this method by extracting text-related information from its description or transcript.

Meta-preferences Classification: To classify a new user’s meta-preferences, that is to mainly distinguish the user’s RP preferences from GP ones (since those two are the most revealed), clustering can be performed based on the click histories of users who clicked

on similar items to the new user (e.g., by measuring the cosine similarity between users’ consumption patterns). The clustering can be performed using at least two types of features: (i) nutrition levels of clicked items; (ii) interaction patterns of users over time (e.g. frequency of interactions, beginning and end points of users’ journeys). K-means clustering, Gaussian Mixture Models, and density-based spatial clustering can be used to perform such clustering (Milecia McGregor, 2020). Note that the clusters are dynamically changing as users interact with more items and the groupings of similar users change.

4.1.2 Give users more control over content selection

Providing users with some macro-level control over recommendations is essential for respecting their autonomy (Djeffal et al., 2022). My system design consists of two aspects of content users can actively control: (i) the proportion of ‘nutritious’ content (or roughly speaking, the proportion of RP content), which I call α ; (ii) how closely the recommended content matches the user’s previous click history, which I call β as the similarity degree. Table 2 shows how adjusting these two parameters could affect a user’s recommendation landscape. Further experiments are required to determine the default values for α and β . For β , the measure of similarity between new items and previously clicked items can be calculated using topic similarity, e.g. the distance between items on a hierarchical taxonomy of topics (Figure 1 shows a sample taxonomy). This approach will inevitably result in some randomness in recommendations within the same topic/subtopic, so I will discuss how to further direct recommendations toward self-realisation using the rules defined in Section 4.1.3.

	Less Nutritious Content	More Nutritious Content
Low Similarity	A novel set of content likely to be dominated by ‘junk food’	A novel set of content that are likely to be thought-evoking
High Similarity	Easily digestible content on existing interests	In-depth understanding on existing interests

Table 2: Potential scenarios based on user’s choices on (i) the proportion of nutritious content (horizontal); (ii) how different the recommended content should be from their existing interests (vertical).

4.1.3 Help users discover their underdeveloped preferences

The key step to self-realisation is supporting users in discovering their underdeveloped tastes. I formulate this step as to ensure that users’ ‘righteous’ (R) preferences take precedence over GP, as we recommender designers don’t want to waste your time on things you know you don’t want to. However, users can always adjust the amount of GP content through α based on their needs.

Evaluation Metrics: To evaluate the success of exploration of underdeveloped tastes, I propose two metrics to be measured in the space of R content: (i) inter-topic diversity, or the variety of new topics explored; (ii) intra-topic depth, or the number of items consumed within the explored topics. There is a natural trade-off between the two metrics due to

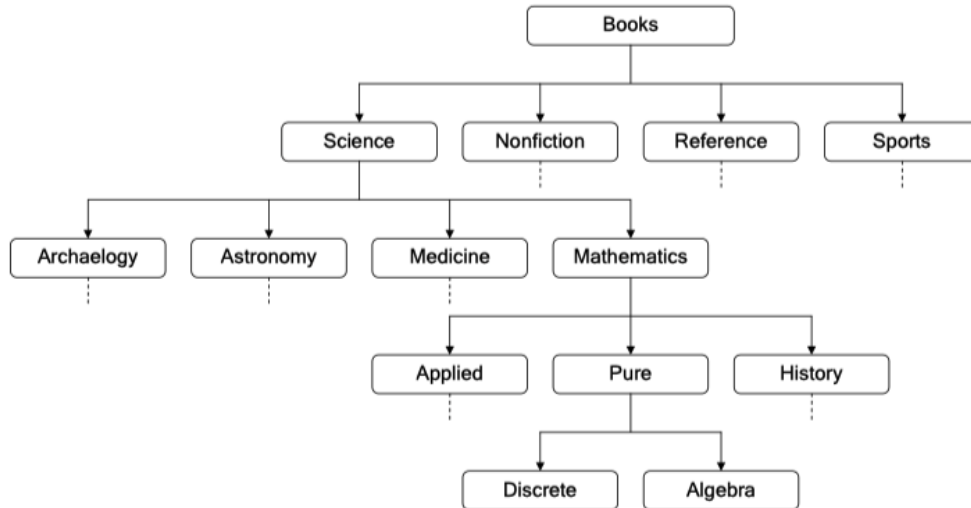


Figure 1: Fragment from the Amazon.com book classification taxonomy (Ziegler et al., 2005)

limited user time, and how to balance them into one unified metric is a question about how to evaluate the success of self-discovery and is open to debate.

The Complete Model: Given the classified meta-preferences (Section 4.1.1), as well as the user-defined α and β (Section 4.1.2), populate the recommendation page for a user with the following strategy: define the proportions of the R and GP content for the user based on α , while diversifying the overall content based on β . To make up for the likely randomness in recommendations, two additional rules are proposed to facilitate the exploration of underdeveloped tastes:

- Encourage user acceptance of new interests: When there is a recommendation movement towards a new ‘nutritious’ topic, prioritise content in the topic that is relatively easier to understand (e.g. low in technical value).
- Take suggestions from ‘taste-based’ groups: Prioritise popular content consumed by other users in the same group as used in the meta-preference classifier, similar to the idea of user-to-user collaborative filtering (Baptiste Rocca, 2021), in which similar users’ preferences are used as reference to the target user.

Algorithmically, the above two rules can be incorporated as constraints into a multi-objective optimisation problem (along with two constraints on α and β), and the optimisation goal is to balance the trade-off between accuracy (accurately predicting a user’s likelihood of clicking an item) and the success of self-realisation (based on the two proposed evaluation metrics). To account for the temporal-changing nature of RS, the optimisation could be performed using reinforcement learning (RL) via a constraint Markov decision process, where the agent learns a long-term policy with dynamic constraints applied at each iteration to optimise for the rewards (Afsar et al., 2021; Liu et al., 2021).

4.2 Datasets and Evaluation

In response to RQ4, there are several static benchmark datasets available for testing recommendation algorithms. As a starting point, I present three datasets focusing on applications of videos, news articles, and books. All the presented datasets contain user interaction histories, the associated time stamps, and item attributes (e.g. topic categories) that are needed for the modelling.

- **MovieLens 25M Dataset**¹: The dataset contains 25 million ratings and 1.1 million tag applications across 62423 movies, created by 162541 users from 1995 to 2019.
- **MIND (Microsoft News Dataset)**²: This dataset was collected from 1 million users who clicked on at least 5 news stories between October 12 and November 22, 2019. In addition, the dataset contains embeddings of relations between entities derived from knowledge graph.
- **Goodreads Book Dataset**³: This dataset contains reviews for over 2 million books and is still being updated. It also includes graphs of similar books, and multiple levels of user interaction, ranging from adding books to a ‘shelf’, rating, and reading.

For testing the algorithms in a dynamic environment, Google’s RecSim (Ie et al., 2019) is a great simulation environment for developing RL algorithms for RS, as it can simulate some aspects of real-world user behaviour and test the robustness of RL algorithms under dynamic user responses.

In addition to quantifiable metrics, we need human feedback on how well the proposed algorithms aid self-realisation. Therefore, a prototype RS application based on the above datasets may need to be developed and tested by real users. For user feed back, paid participants could be recruited via Amazon Mechanical Turk to complete a questionnaire regarding recommendation quality and diversity, taste coverage, and taste development potential. (Wilkinson, 2018).

5 Limitations

While there are many finer technical details to work out in this proposal, I would like to point out its two major conceptual limitations: (i) By promoting self-realisation, we might in fact widen rather than close the gap between diversity seekers and challenge avoiders, resulting in different levels of autonomy protection for users (Mattis et al., 2022); (ii) The new recommender design has the potential to discriminate against certain user interests, and the degree to which the RS can determine what is good or bad needs to be considered carefully.

¹<https://grouplens.org/datasets/movielens/>

²<https://msnews.github.io/>

³<https://sites.google.com/eng.ucsd.edu/ucsdbookgraph/home?pli=1>

References

- Afsar, M. M., Crump, T., & Far, B. (2021). Reinforcement learning based recommender systems: A survey. <https://doi.org/10.48550/ARXIV.2101.06286>
- Amatriain, X., Pujol, J. M., Tintarev, N., & Oliver, N. (2009). Rate it again: Increasing recommendation accuracy by user re-rating. *ACM Conference on Recommender Systems*.
- Aragoncillo Caballero, L., & Orús, C. (2018). Impulse buying behaviour: An online-offline comparative and the impact of social media: Comportamiento de compra impulsiva: Comparativa online-offline e impacto de las redes sociales. *Spanish Journal of Marketing - ESIC*, 22. <https://doi.org/10.1108/SJME-03-2018-007>
- Ashton, H., & Franklin, M. (2022a). The problem of behaviour and preference manipulation in ai systems.
- Ashton, H., & Franklin, M. (2022b). Solutions to preference manipulation in recommender systems require knowledge of meta-preferences. <https://doi.org/10.48550/ARXIV.2209.11801>
- Baptiste Rocca. (2021). *Introduction to recommender systems*. <https://towardsdatascience.com/introduction-to-recommender-systems-6c66cf15ada> (accessed: 02.01.2022)
- Benn, S. I. (1988). Self-realization, instinctual freedom, and autonomy. In *A theory of freedom* (pp. 199–212). Cambridge University Press. <https://doi.org/10.1017/CBO9780511609114.013>
- Botes, W. (2022). Autonomy and the social dilemma of online manipulative behavior. *AI and Ethics*. <https://doi.org/10.1007/s43681-022-00157-5>
- Boyd, D. (2010). Streams of content, limited attention: The flow of information through social media.
- Calvo, R. A., Peters, D., Vold, K., & Ryan, R. M. (2020). Supporting human autonomy in ai systems: A framework for ethical enquiry. *Philosophical Studies Series*.
- Cosley, Lam, K, S., Tsibouklis, J., Albert, Istvan, Konstan, A, J., & Riedl. (2003). Is seeing believing?: How recommender system interfaces affect users’ opinions. <https://doi.org/10.1145/642611.642713>
- Dirnhuber, J. (2019). ‘INSTAGRAM HELPED KILL MY GIRL’ Heartbroken dad claims Instagram ‘helped to kill his 14-year-old daughter’ who took her own life after viewing suicide posts. <https://www.thesun.co.uk/news/8258105/ian-russell-molly-instagram-killed-daughter/> (accessed: 14.12.2022)
- Djeffal, C., Magrani, E., & Hitrova, C. (2022). Recommender systems and autonomy: A role for regulation of design, rights, and transparency.
- Doyle, J. (2012). Survey of time preference, delay discounting models. *Judgment and Decision Making*, 8. <https://doi.org/10.2139/ssrn.1685861>
- Dumbrava, C. (2018). *Key social media risks to democracy: Risks from surveillance, personalisation, disinformation, moderation and microtargeting*. [https://www.europarl.europa.eu/RegData/etudes/IDAN/2021/698845/EPRS_IDA\(2021\)698845_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/IDAN/2021/698845/EPRS_IDA(2021)698845_EN.pdf) (accessed: 09.12.2022)
- Evans, M. (1998). Self-realization. <https://doi.org/10.4324/9780415249126-L091-1>
- Fogg, B. (2003). *Persuasive technology: Using computers to change what we think and do*. Morgan Kaufmann Publishers Inc.

- Gretzel, U., & Fesenmaier, D. R. (2006). Persuasion in recommender systems. *International Journal of Electronic Commerce*, 11(2), 81–100. <https://doi.org/10.2753/JEC1086-4415110204>
- Hinds, J., Williams, E. J., & Joinson, A. N. (2020). “it wouldn’t happen to me”: Privacy concerns and perspectives following the cambridge analytica scandal. *International Journal of Human-Computer Studies*, 143, 102498. <https://doi.org/https://doi.org/10.1016/j.ijhcs.2020.102498>
- Hiniker, A., Hong, S. (, Kohno, T., & Kientz, J. A. (2016). Mytime: Designing and evaluating an intervention for smartphone non-use. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 4746–4757. <https://doi.org/10.1145/2858036.2858403>
- Ie, E., Hsu, C.-w., Mladenov, M., Jain, V., Narvekar, S., Wang, J., Wu, R., & Boutilier, C. (2019). Recsim: A configurable simulation platform for recommender systems. <https://doi.org/10.48550/ARXIV.1909.04847>
- John Walter Bordages, J. (1989). Self-actualization and personal autonomy. *Psychological Reports*, 64(3_suppl), 1263–1266. <https://doi.org/10.2466/pr0.1989.64.3c.1263>
- Kleingeld, P. (2017). The principle of autonomy in kant’s moral theory: Its rise and fall. In E. Watkins (Ed.), *Kant on persons and agency* (pp. 61–80). Cambridge University Press. <https://doi.org/10.1017/9781316856529.005>
- Knijnenburg, B. P., Sivakumar, S., & Wilkinson, D. (2016). Recommender systems for self-actualization. *Proceedings of the 10th ACM Conference on Recommender Systems*, 11–14. <https://doi.org/10.1145/2959100.2959189>
- Kurdi, M. (2019). Measuring content complexity of technical texts: Machine learning experiments. https://doi.org/10.1007/978-3-030-23207-8_28
- Liu, W., Liu, F., Tang, R., Liao, B., Chen, G., & Heng, P. A. (2021). Balancing accuracy and fairness for interactive recommendation with reinforcement learning. <https://doi.org/10.48550/ARXIV.2106.13386>
- Mattis, N., Masur, P., Möller, J., & van Atteveldt, W. (2022). Nudging towards news diversity: A theoretical framework for facilitating diverse news consumption through recommender design. *New Media & Society*, 0(0), 14614448221104413. <https://doi.org/10.1177/14614448221104413>
- Michael Schrage. (2020). *Recommendation engines: data-driven self-help is the future of the internet*. <https://blogs.lse.ac.uk/businessreview/2020/10/13/recommendation-engines-data-driven-self-help-is-the-future-of-the-internet/> (accessed: 27.12.2022)
- Milano, S., Taddeo, M., & Floridi, L. (2020a). Ethical Aspects of Multi-stakeholder Recommendation Systems. *The Information Society*, 37, 1–11. <https://doi.org/10.1080/01972243.2020.1832636>
- Milano, S., Taddeo, M., & Floridi, L. (2020b). Recommender Systems and Their Ethical Challenges. *AI Soc.*, 35(4), 957–967. <https://doi.org/10.1007/s00146-020-00950-y>
- Milecia McGregor. (2020). *8 Clustering Algorithms in Machine Learning that All Data Scientists Should Know*. <https://www.freecodecamp.org/news/8-clustering-algorithms-in-machine-learning-that-all-data-scientists-should-know/> (accessed: 05.01.2022)
- Natali Helberger. (2020). *Challenging rabbit holes: towards more diversity in news recommendation systems*. <https://blogs.lse.ac.uk/medialse/2020/07/02/challenging->

- [rabbit-holes-towards-more-diversity-in-news-recommendation-systems/](#) (accessed: 23.12.2022)
- Nguyen, T. T., Hui, P.-M., Harper, F. M., Terveen, L., & Konstan, J. A. (2014). Exploring the filter bubble: The effect of using recommender systems on content diversity. *Proceedings of the 23rd International Conference on World Wide Web*, 677–686. <https://doi.org/10.1145/2566486.2568012>
- Pariser, E. (2012). The filter bubble: How the new personalized web is changing what we read and how we think.
- Susser, D., Roessler, B., & Nissenbaum, H. (2019). Technology, autonomy, and manipulation [Publisher Copyright: © 2019, Alexander von Humboldt Institute for Internet and Society. All rights reserved.]. *Internet Policy Review*, 8(2). <https://doi.org/10.14763/2019.2.1410>
- Wilkinson, D. (2018). Testing a recommender system for self-actualization. *Proceedings of the 12th ACM Conference on Recommender Systems*, 543–547. <https://doi.org/10.1145/3240323.3240324>
- Ziegler, C.-N., McNee, S. M., Konstan, J. A., & Lausen, G. (2005). Improving recommendation lists through topic diversification. *Proceedings of the 14th International Conference on World Wide Web*, 22–32. <https://doi.org/10.1145/1060745.1060754>